Setup the “First Notch Filter”

Goal : Filter out the lift-motor vibration peak (and its harmonics) so the attitude controller isn’t fighting noise; especially in Q-modes.

(mechanical fixes → logging → FFT → set notches → verify → tune)

Quick summary (what we’ll do)

1.Fix the obvious mechanical vibration sources first (props, motors, mountings).

2.Collect high-rate IMU data (raw IMU or IMU batch sampling) while the aircraft hovers.

3.Use ArduPilot’s Filter/FFT tools to find the dominant gyro frequency and bandwidth.

4.Enable and configure the **Harmonic Notch** (INS_HNTCH_*) using the measured values.

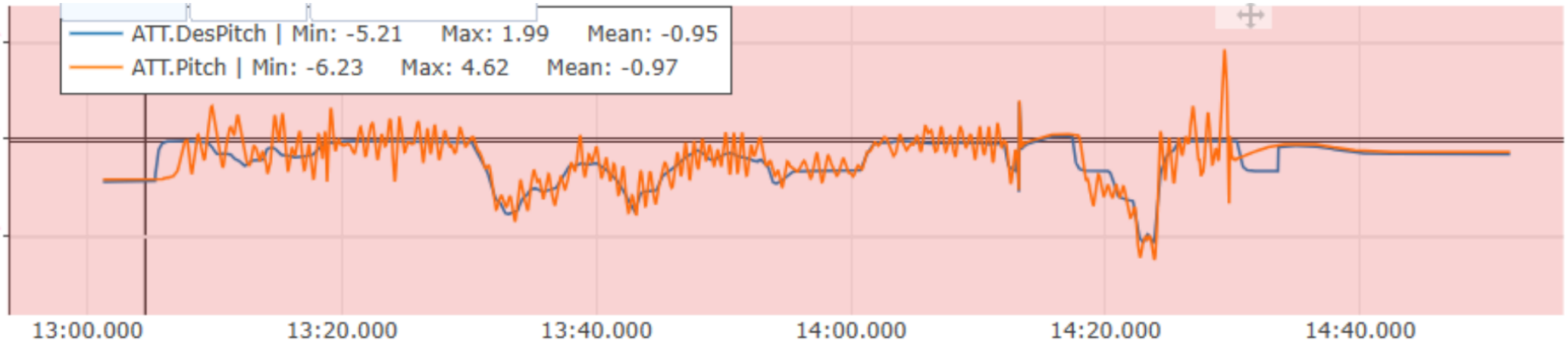
5.Re-test, verify the FFT / Vibe graphs, then relax or raise gyro low-pass (INS_GYRO_FILTER) and finish PID tuning.

(ArduPilot docs with full details: see the Notch/QuadPlane tuning pages below.) Managing Gyro Noise with the Dynamic Harmonic Notch Filters — Plane documentation

Edited portfolio excerpt. No raw logs, location maps or full parameter files are distributed.

VTOL flight-log analysis

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